Quantum Computing, Quantum Machine Learning and Quantum Information Theories

Morten Hjorth-Jensen[1]

Department of Physics, University of Oslo, Norway[1]

February 4, 2026

Plans for the week of February 2-6, 2026

1. Density matrices, Entanglement, Bell states and entropy
2. Getting started with gates and more
3. For exercises, see end of these slides

Motivation

In order to study entanglement and why it is so important for quantum computing, we need to introduce some basic measures and useful quantities. For these endeavors, we will use our two-qubit system from the second lecture in order to introduce and repeat, through examples, density matrices and entropy. In particular we will focus on the so-called Bell states discussed in the previous lecture.
These two quantities, together with technicalities like the Schmidt decomposition define important quantities in analyzing quantum computing examples.
The Schmidt decomposition is again a linear decomposition which allows us to express a vector in terms of tensor product of two inner product spaces. In quantum information theory and quantum computing it is widely used as away to define and describe entanglement.

Reminder on density matrices and traces

We have the spectral decomposition of a given operator $\boldsymbol{A}$ given by

$$\boldsymbol{A} = \sum_i \lambda_i |i\rangle\langle i|,$$

with the ONB $|i\rangle$ being eigenvectors of $\boldsymbol{A}$ and λ_i being the eigenvalues. Similarly, a operator which is a function of $\boldsymbol{A}$ is given by

$$f(\boldsymbol{A}) = \sum_i f(\boldsymbol{A})|i\rangle\langle i|.$$

The trace of a product of matrices is cyclic, that is

$$\mathrm{tr}[\boldsymbol{ABC}]) = \mathrm{tr}[\boldsymbol{BCA}]) = \mathrm{tr}[\boldsymbol{CBA}]),$$

and we have also

$$\mathrm{tr}[\boldsymbol{A}|\psi\rangle\langle\psi|]) = \langle\psi|\boldsymbol{A}|\psi\rangle.$$

Definition of density matrix

Using the spectral decomposition we defined also the density matrix as

$$\rho = \sum_i p_i |i\rangle\langle i|,$$

where the probability p_i are the eigenvalues of the density linked with the ONB $|i\rangle$.

The trace of the density matrix is $\mathrm{tr}\rho = 1$ and it is invariant under unitary transformations $|\psi_i'\rangle = \boldsymbol{U}|\psi_i\rangle$. The unitary transformation of the density matrix gives, with $\boldsymbol{U}^\dagger\boldsymbol{U} = \boldsymbol{U}^T\boldsymbol{U} = \boldsymbol{I}$,

$$\boldsymbol{U}\rho\boldsymbol{U}^\dagger = \sum_i pi_i \boldsymbol{U}|\psi_i\rangle\langle\psi_i|\boldsymbol{U}^\dagger,$$

and with the unitary transformation it is easy to show that thet trace of the transformaed density matrix is equal to one,

$$\mathrm{tr}\left[\boldsymbol{U}\rho\boldsymbol{U}^\dagger\right] = \mathrm{tr}\left[\boldsymbol{U}\boldsymbol{U}^\dagger\rho\right] = 1.$$

Revisiting our Bell states and our entanglement encounter, two qubit system

We define a system that can be thought of as composed of two subsystems A and B. Each subsystem has computational basis states

$$|0\rangle_{\mathrm{A,B}} = \begin{bmatrix} 1 & 0 \end{bmatrix}^T \qquad |1\rangle_{\mathrm{A,B}} = \begin{bmatrix} 0 & 1 \end{bmatrix}^T.$$

The subsystems could represent single particles or composite many-particle systems of a given symmetry.

Computational basis

This leads to the many-body computational basis states

$$|00\rangle = |0\rangle_\mathrm{A} \otimes |0\rangle_\mathrm{B} = \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix}^T,$$

and

$$|01\rangle = |0\rangle_\mathrm{A} \otimes |1\rangle_\mathrm{B} = \begin{bmatrix} 0 & 1 & 0 & 0 \end{bmatrix}^T,$$

and

$$|10\rangle = |1\rangle_\mathrm{A} \otimes |0\rangle_\mathrm{B} = \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix}^T,$$

and finally

$$|11\rangle = |1\rangle_\mathrm{A} \otimes |1\rangle_\mathrm{B} = \begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix}^T.$$

Bell states

The above computational basis states, which define an ONB, can in turn be used to define another ONB. As an example, consider the so-called Bell states

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}\left[|00\rangle + |11\rangle\right] = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\0\\0\\1\end{bmatrix},$$

$$|\Phi^-\rangle = \frac{1}{\sqrt{2}}\left[|00\rangle - |11\rangle\right] = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\0\\0\\-1\end{bmatrix},$$

The next two

$$|\Psi^{+}\rangle = \frac{1}{\sqrt{2}}\left[|10\rangle + |01\rangle\right] = \frac{1}{\sqrt{2}}\begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix},$$

and

$$|\Psi^{-}\rangle = \frac{1}{\sqrt{2}}\left[|10\rangle - |01\rangle\right] = \frac{1}{\sqrt{2}}\begin{bmatrix} 0 \\ 1 \\ -1 \\ 0 \end{bmatrix}.$$

It is easy to convince oneself that these states also form an orthonormal basis.

Measurement

Measuring one of the qubits of one of the above Bell states, automatically determines, as we will see below, the state of the second qubit. To convince ourselves about this, let us assume we perform a measurement on the qubit in system A by introducing the projections with outcomes 0 or 1 as

$$\boldsymbol{P}_0 = |0\rangle\langle 0|_A \otimes \boldsymbol{I}_B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

for the projection of the $|0\rangle$ state in system A and similarly

$$\boldsymbol{P}_1 = |1\rangle\langle 1|_A \otimes \boldsymbol{I}_B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \otimes \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

for the projection of the $|1\rangle$ state in system A.

Probability of outcome

We can then calculate the probability for the various outcomes by computing for example the probability for measuring qubit 0

$$\langle \Phi^+ | \boldsymbol{P}_0 | \Phi^+ \rangle = \frac{1}{2} \left[\langle 00| + \langle 11| \right] \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \left[|00\rangle + |11\rangle \right] = \frac{1}{2}.$$

Similarly, we obtain

$$\langle \Phi^+ | \boldsymbol{P}_1 | \Phi^+ \rangle = \frac{1}{2} \left[\langle 00| + \langle 11| \right] \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \left[|00\rangle + |11\rangle \right] = \frac{1}{2}.$$

States after measurement

After the above measurements the system is in the states

$$|\Phi'_0\rangle = \sqrt{2}\,[|0\rangle\langle 0|_A \otimes \boldsymbol{I}_B]\,|\Phi^+\rangle = |00\rangle,$$

and

$$|\Phi'_1\rangle = \sqrt{2}\,[|1\rangle\langle 1|_A \otimes \boldsymbol{I}_B]\,|\Phi^+\rangle = |11\rangle.$$

We see from the last two equations that the state of the second qubit is determined even though the measurement has only taken place locally on system A.

Other states

If we on the other hand consider a state like

$$|00\rangle = |0\rangle_A \otimes |0\rangle_B,$$

this is a pure **product** state of the single-qubit, or single-particle states, of two qubits (particles) in system A and system B, respectively. We call such a state for a **pure product state**. Quantum states that cannot be written as a mixture of other states are called pure quantum states or just product states, while all other states are called mixed quantum states.

More on Bell states

A state like one of the Bell states (where we introduce the subscript AB to indicate that the state is composed of single states from two subsystem)

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}\left[|00\rangle_{AB} + |11\rangle_{AB}\right],$$

is on the other hand a mixed state and we cannot determine whether system A is in a state 0 or 1. The above state is a superposition of the states $|00\rangle_{AB}$ and $|11\rangle_{AB}$ and it is not possible to determine individual states of systems A and B, respectively.

Entanglement

We say that the state is entangled. This yields the following definition of entangled states: a pure bipartite state $|\psi\rangle_{AB}$ is entangled if it cannot be written as a product state $|\psi\rangle_A \otimes |\phi\rangle_B$ for any choice of the states $|\psi\rangle_A$ and $|\phi\rangle_B$. Otherwise we say the state is separable.

Maximally entangled

A so-called maximally entangled state for a bipartite system has equal probability amplitudes

$$|\Psi\rangle = \frac{1}{\sqrt{d}} \sum_{i=0}^{d-1} |ii\rangle.$$

We call a bipartite state composed of systems A and B (these systems can be single-particle systems, or single-qubit systems representing low-lying states of complicated many-body systems) for separable if its density matrix ρ_{AB} can be written out as the tensor product of the individual density matrices ρ_A and ρ_B, that is we have for a given probability distribution p_i

$$\rho_{AB} = \sum_i p_i \rho_A(i) \otimes \rho_B(i).$$

Summarizing the four Bell states (two qubits)

Let $\{|0\rangle, |1\rangle\}$ be the computational basis for a single qubit. For two qubits, the computational basis is $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$.

The **Bell states** are the four maximally entangled two-qubit states

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}\left(|00\rangle + |11\rangle\right), \qquad |\Phi^-\rangle = \frac{1}{\sqrt{2}}\left(|00\rangle - |11\rangle\right), \tag{1}$$

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle + |10\rangle\right), \qquad |\Psi^-\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle - |10\rangle\right). \tag{2}$$

They form an orthonormal basis of $\mathbb{C}^2 \otimes \mathbb{C}^2$ and are eigenstates of correlated observables (parity and certain two-qubit Pauli products).

Density matrix for all four Bell states

For a **pure** state $|\psi\rangle$, the density operator is

$$\rho = |\psi\rangle\langle\psi|.$$

Example:

$$\rho_{\Phi^+} = \frac{1}{2}\big(|00\rangle\langle 00| + |00\rangle\langle 11| + |11\rangle\langle 00| + |11\rangle\langle 11|\big).$$

Similarly, expanding each Bell projector

$$\rho_{\Phi^-} = \frac{1}{2}\big(|00\rangle\langle 00| - |00\rangle\langle 11| - |11\rangle\langle 00| + |11\rangle\langle 11|\big), \qquad (3)$$

$$\rho_{\Psi^+} = \frac{1}{2}\big(|01\rangle\langle 01| + |01\rangle\langle 10| + |10\rangle\langle 01| + |10\rangle\langle 10|\big), \qquad (4)$$

$$\rho_{\Psi^-} = \frac{1}{2}\big(|01\rangle\langle 01| - |01\rangle\langle 10| - |10\rangle\langle 01| + |10\rangle\langle 10|\big). \qquad (5)$$

Matrix forms in the computational basis

Use ordering $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$. Then

$$\rho_{\Phi^+} = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}, \qquad \rho_{\Phi^-} = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 1 \end{pmatrix}$$

More density matrices

$$\rho_{\Psi^+} = \frac{1}{2}\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \qquad \rho_{\Psi^-} = \frac{1}{2}\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Proof that the Bell states form an orthonormal basis

To show they form an orthonormal basis it suffices to prove first
Normalization: $\langle \Phi^{\pm} | \Phi^{\pm} \rangle = 1$ and $\langle \Psi^{\pm} | \Psi^{\pm} \rangle = 1$. For example,

$$\langle \Phi^{+} | \Phi^{+} \rangle = \frac{1}{2}\big(\langle 00|00\rangle + \langle 00|11\rangle + \langle 11|00\rangle + \langle 11|11\rangle\big) = \frac{1}{2}(1+0+0+1) = 1.$$

Orthogonality

Inner products between distinct Bell states vanish. Example:

$$\langle \Phi^+ | \Phi^- \rangle = \frac{1}{2}\big(\langle 00|00\rangle - \langle 00|11\rangle + \langle 11|00\rangle - \langle 11|11\rangle\big) = \frac{1}{2}(1-0+0-1) = 0.$$

We leave the other states as exercise. Completeness: Since there are four orthonormal vectors in a four-dimensional space $\Rightarrow$ they form a basis.

Pure and mixed states (definitions)

Density operator: a quantum state can be represented by a positive semidefinite operator ρ with $\mathrm{Tr}(\rho) = 1$.

Pure state:

$$\rho \text{ is pure} \iff \rho = |\psi\rangle\langle\psi| \iff \rho^2 = \rho \iff \mathrm{Tr}(\rho^2) = 1.$$

Mixed state: a probabilistic ensemble of pure states $(p_i, |\psi_i\rangle)$,

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|, \qquad p_i \geq 0, \quad \sum_i p_i = 1.$$

Then typically $\rho^2 \neq \rho$ and

$$\mathrm{Tr}(\rho^2) < 1.$$

Example: maximally mixed single-qubit state $\rho = \frac{I_2}{2}$ has $\mathrm{Tr}(\rho^2) = \frac{1}{2}$.

Reduced density matrix via partial trace (full details)

Let ρ_{AB} be a two-qubit density matrix on $\mathcal{H}_A \otimes \mathcal{H}_B$. The reduced state of qubit A is

$$\rho_A = \mathrm{Tr}_B(\rho_{AB}) = \sum_{b\in\{0,1\}} (I_A \otimes \langle b|)\, \rho_{AB}\, (I_A \otimes |b\rangle)\,.$$

Worked example for $|\Phi^+\rangle$:

$$\rho_{\Phi^+} = \frac{1}{2}\Big(|00\rangle\langle 00| + |00\rangle\langle 11| + |11\rangle\langle 00| + |11\rangle\langle 11|\Big).$$

Take Tr_B term-by-term, using $\mathrm{Tr}_B\big(|ab\rangle\langle cd|\big) = \langle d|b\rangle\, |a\rangle\langle c|$:

$$\mathrm{Tr}_B(|00\rangle\langle 00|) = \langle 0|0\rangle|0\rangle\langle 0| = |0\rangle\langle 0|, \tag{6}$$
$$\mathrm{Tr}_B(|00\rangle\langle 11|) = \langle 1|0\rangle|0\rangle\langle 1| = 0, \tag{7}$$
$$\mathrm{Tr}_B(|11\rangle\langle 00|) = \langle 0|1\rangle|1\rangle\langle 0| = 0, \tag{8}$$
$$\mathrm{Tr}_B(|11\rangle\langle 11|) = \langle 1|1\rangle|1\rangle\langle 1| = |1\rangle\langle 1|. \tag{9}$$

More derivations

Therefore,

$$\rho_A = \mathrm{Tr}_B(\rho_{\Phi^+}) = \frac{1}{2}\left(|0\rangle\langle 0| + |1\rangle\langle 1|\right) = \frac{I_2}{2}.$$

Same result holds for all four Bell states: each reduced state is maximally mixed. See also exercises below.

Measurement on one qubit (projective measurement)

Consider measuring qubit A in the computational basis $\{|0\rangle, |1\rangle\}$.
Projectors on the two-qubit space:

$$P_0 = (|0\rangle\langle 0|) \otimes I, \qquad P_1 = (|1\rangle\langle 1|) \otimes I.$$

For a pre-measurement state ρ_{AB},

$$p(0) = \mathrm{Tr}(P_0 \rho_{AB}), \qquad p(1) = \mathrm{Tr}(P_1 \rho_{AB}), \qquad p(0) + p(1) = 1.$$

Post-measurement (collapsed) state conditioned on outcome $k \in \{0, 1\}$:

$$\rho_{AB}^{(k)} = \frac{P_k \rho_{AB} P_k}{p(k)}.$$

Example

Example: for $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$, one finds

$$p(0) = \frac{1}{2}, \quad p(1) = \frac{1}{2},$$

and

$$\rho_{AB}^{(0)} = |00\rangle\langle 00|, \qquad \rho_{AB}^{(1)} = |11\rangle\langle 11|.$$

Thus measuring qubit A collapses qubit B to the same outcome (perfect correlations).

Entropies and density matrices

We discuss first the classical information entropy. Thereafter we define the quantum-mechanical quantity, commonly known as the Von-Neumann entropy.

Classical entropy

Entropy and basic ddea

Entropy quantifies **uncertainty** or **information content**. For a single event of probability p, the information content is

$$I(p) = -\log_2 p.$$

Rare events ($p \to 0$) carry large information; certain events ($p = 1$) carry zero information.

Shannon entropy

For a discrete random variable X with outcomes x:

$$H(X) = -\sum_{x} p(x) \log_2 p(x).$$

Binary entropy (outcomes with probabilities p and $1 - p$):

$$H_2(p) = -p \log_2 p - (1 - p) \log_2(1 - p).$$

This leads to $H_2(0) = H_2(1) = 0$ (max certainty) and $H_2(1/2) = 1$ (maximum uncertainty).

Binary Shannon entropy curve

Compute $H_2(p)$ for representative p values.

```
import math

def H2(p):
    eps = 1e-15
    p = min(max(p, eps), 1.0-eps)
    return -p*math.log(p, 2) - (1-p)*math.log(1-p, 2)

for p in [0.0, 0.1, 0.25, 0.5, 0.75, 0.9, 1.0]:
    # clamp for endpoints
    pp = min(max(p, 1e-15), 1-1e-15)
    print(p, H2(pp))
```

Shannon information entropy

We start our discussions with the classical information entropy, or just Shannon entropy, before we move over to a quantum mechanical way to define the entropy based on the density matrices discussed earlier.
We define a set of random variables $X = \{x_0, x_1, \dots, x_{n-1}\}$ with probability for an outcome $x \in X$ given by $p_X(x)$, the classical information entropy is defined as

$$S = -\sum_{x \in X} p_X(x) \log_2 p_X(x).$$

Why this expression? What does it mean?

Mathematics of entropy

What is the basic idea of the entropy as it is used in information theory (we leave out the standard description from statistical physics here)?
We want to have a measure of unlikelihood. Consider a simple binary system, with two outcomes, true and false with true given by a probability p and false given by $1-p$. Since p represents a probability, and assuming the probabilities are properly normalized, we have $p \in [0, 1]$.
Suppose know that we know precisely p, for example we could set $p = 1$. This means that there is only one outcome, namely the true one and false is equal to zero. We know thus (exactly) which state the system is in. If we set $p = 0$, then we know that the outcome is false, without doubt. Can we find a way to, through a specific function to say with a given certainty that the system is in a specific state? Let us call this function for S.

More on the mathematics of entropy

We could for example use

$$S(x) = -\log_2 p(x),$$

as potential function. Note that if we change the log-base, we only change the results by a scaling constant.
When $p(x) = 1$, then $S(x) = 0$ and we could say that the probability of being in one specific state is uniquely defined since S is then zero. This function is also continuous, it is additive (which is very useful of we have independent and identically distributed stochastic variables), it is also high for unlikely events since when p goes to zero (unlikely event) it becomes very large. It is also a non-negative function.

Changing the expression

We note however that if we go back to our binary model and would like to describe the entropy in terms of the variable p, we end up with p being zero for the false outcome. The latter has probability one ($p = 0$). Using that

$$\lim_{x \to -\infty} -x \log_2 x = 0,$$

we can define the classical information entropy as

$$S(x) = -p(x) \log_2 p(x).$$

If we now consider a series of stochastic variables $X = \{x_0, x_1, \dots, x_{n-1}\}$ with probability for an outcome $x \in X$ given by $p_X(x)$, the entropy becomes

$$S_X = -\sum_{x \in X} p_X(x_i) \log_2 p_X(x_i).$$

Binary example

If we now use the above binary example, we have (using that $p_1 = p$ and $p_2 = 1 - p$)

$$S = -\sum_{i=1}^{2} p_i \log_2 p_i = -p \log_2 p - (1-p) \log_2 (1-p).$$

For $p = 0$ we have $S = 0$ and for $p = 1$ we get $S = 0$. Thus, for the two uniquely defined states we have entropy zero and a precise knowledge of the state. The maximum value is at $p = 0.5$ and this is a situation where our level of knowledge is no better than us tossing a coin. This simple example demonstrates how we can use the above mathematical expression as a way to say whether we have a well-defined outcome or not. Our lack of knowledge is expressed by the largest entropy value.

Quantum entropy

A quantum state can be:

- **Pure:** $\rho = |\psi\rangle\langle\psi|$, with $\rho^2 = \rho$
- **Mixed:** $\rho = \sum_j p_j |\psi_j\rangle\langle\psi_j|$, with $\rho^2 \neq \rho$

Properties:

$$\rho^\dagger = \rho, \quad \rho \succeq 0, \quad \mathrm{Tr}(\rho) = 1.$$

Von Neumann entropy

Quantum analogue of Shannon entropy:

$$S(\rho) = -\mathrm{Tr}(\rho \log_2 \rho)\,.$$

If ρ has eigenvalues $\{\lambda_i\}$:

$$S(\rho) = -\sum_i \lambda_i \log_2 \lambda_i.$$

- Pure state $\Rightarrow \{\lambda\} = \{1, 0, \dots\} \Rightarrow S(\rho) = 0$
- Maximally mixed in d dimensions: $\rho = \frac{I_d}{d} \Rightarrow S(\rho) = \log_2 d$

Von Neumann entropy

Using the density matrix ρ, we define the quantum mechanical equivalent of the classical entropy as

$$S = -\mathrm{Tr}[\rho \log_2 \rho].$$

This is the so-called Von Neumann entropy. How did we arrive at this expression?
And why can't we use the classical definition for a quantum mechanical system as well?

The density matrix again

Let us assume we are studying specific system A. The density matrix, using an ONB basis ψ_j, is defined as

$$\boldsymbol{\rho}_A = \sum_j \lambda_j |\psi_j\rangle\langle\psi_j| = \sum_j p_j |\psi_j\rangle\langle\psi_j|,$$

and the eigenvalues λ_j are the probabilities (or overlap coefficients) of being in a specific state ψ_j.
Note that the density matrix/operator is a semi-positive definite matrix.

Linking with a new expression for the entropy

Using a unitary transformation $\boldsymbol{U}$ we can transform the density matrix into a diagonal matrix $\boldsymbol{D}_a$ where the eigenvalues are the above mentioned probabilities (overlap coefficients squared). We can then define

$$\boldsymbol{D}_A = \boldsymbol{U}^\dagger \boldsymbol{\rho}_A \boldsymbol{U} = \begin{bmatrix} p_0 & 0 & 0 & \dots & 0 \\ 0 & p_1 & 0 & \dots & 0 \\ 0 & 0 & p_2 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & 0 & p_{n-1} \end{bmatrix}.$$

Von Neumann entropy

If we now construct the following object

$$\boldsymbol{\rho}_A \log_2 \boldsymbol{\rho}_A,$$

and perform a unitary transformation, recalling that $\boldsymbol{U}\boldsymbol{U}^\dagger = \boldsymbol{U}^\dagger\boldsymbol{U} = \boldsymbol{I}$,

$$\boldsymbol{U}^\dagger \boldsymbol{\rho}_A \log_2 \boldsymbol{\rho}_A \boldsymbol{U} = \boldsymbol{U}^\dagger \boldsymbol{\rho}_A \boldsymbol{I} \log_2 \boldsymbol{\rho}_A \boldsymbol{U} = \boldsymbol{D}_A \boldsymbol{U}^\dagger \log_2 \boldsymbol{\rho}_A \boldsymbol{U},$$

and defining

$$\boldsymbol{U}^\dagger \log_2 \boldsymbol{\rho}_A \boldsymbol{U} = \log_2 \boldsymbol{D}_A,$$

we have that

$$\boldsymbol{U}^\dagger \boldsymbol{\rho}_A \log_2 \boldsymbol{\rho}_A \boldsymbol{U} = \boldsymbol{D}_A \log_2 \boldsymbol{D}_A.$$

$\log_2 \boldsymbol{D}_A$

We have that

$$\log_2 \boldsymbol{D}_A = \begin{bmatrix} \log_2 p_0 & 0 & 0 & \dots & 0 \\ 0 & \log_2 p_1 & 0 & \dots & 0 \\ 0 & 0 & \log_2 p_2 & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & 0 & \log_2 p_{n-1} \end{bmatrix}.$$

Entropy and traces

If we take the trace of

$$\mathrm{Tr}[\boldsymbol{U}^\dagger \boldsymbol{\rho}_A \log_2 \boldsymbol{\rho}_A \boldsymbol{U}] = \mathrm{Tr}[\boldsymbol{D}_A \log_2 \boldsymbol{D}_A],$$

which is nothing but

$$\mathrm{Tr}[\boldsymbol{D}_A \log_2 \boldsymbol{D}_A] = \sum_i p_i \log_2 p_i,$$

which allows us to define the general Von-Neumann entropy as

$$S_A = -\mathrm{Tr}[\boldsymbol{\rho}_A \log_2 \boldsymbol{\rho}_A].$$

Performing a unitary transformation gives the expression we know from classical information theory.

Pure-Python code: partial trace + von Neumann entropy

Goal: compute $\rho_A = \mathrm{Tr}_B(\rho_{AB})$ and

$$S(\rho_A) = -\mathrm{Tr}(\rho_A \log_2 \rho_A) = -\sum_{i=1}^{2} \lambda_i \log_2 \lambda_i,$$

where $\{\lambda_i\}$ are eigenvalues of the 2×2 matrix ρ_A.

```
import math
import cmath

def outer(v):
    """Outer product |v><v| for a statevector v (length 4)."""
    n = len(v)
    return [[v[i] * v[j].conjugate() for j in range(n)] for i in range

def partial_trace_B(rho):
    """
    Partial trace over qubit B for a 2-qubit density matrix rho (4x4).
    Basis ordering: |00>,|01>,|10>,|11>.
    Returns rho_A (2x2).
    """
    rhoA = [[0j, 0j],[0j, 0j]]
    # Indices: |a b> mapped to idx = 2*a + b
    for a in (0,1):
        for c in (0,1):
            s = 0j
            for b in (0,1):
                i = 2*a + b
```

Pure-Python code: measurement on one qubit

We implement projective measurement of qubit A in $\{|0\rangle, |1\rangle\}$.
For a pure state vector $|\psi\rangle$, outcome probabilities are

$$p(k) = \langle\psi|P_k|\psi\rangle, \quad P_k = (|k\rangle\langle k|) \otimes I,$$

and the post-measurement state is

$$|\psi_k\rangle = \frac{P_k|\psi\rangle}{\sqrt{p(k)}}.$$

Below: compute $p(0), p(1)$ and the conditional post-measurement statevectors.

Code (standard Python libraries only)

```
import math

def measure_qubit_A_statevector(psi, outcome):
    """
    Measure qubit A in computational basis on a 2-qubit statevector ps
    outcome = 0 or 1.
    Returns (p, psi_post) where psi_post is normalized statevector.
    """
    # Projector keeps basis states with A=outcome:
    # indices 2*outcome + b for b in {0,1}
    keep = [2*outcome + 0, 2*outcome + 1]
    p = sum((psi[i].conjugate() * psi[i]).real for i in keep)
    if p <= 0.0:
        return 0.0, [0j,0j,0j,0j]
    norm = 1.0 / math.sqrt(p)
    psi_post = [0j,0j,0j,0j]
    for i in keep:
        psi_post[i] = psi[i] * norm
    return p, psi_post

# Example: |Phi+>
inv_sqrt2 = 1.0 / math.sqrt(2.0)
psi = [inv_sqrt2, 0j, 0j, inv_sqrt2]

p0, psi0 = measure_qubit_A_statevector(psi, 0)
p1, psi1 = measure_qubit_A_statevector(psi, 1)

print("p(0) =", p0, "post state =", psi0)
print("p(1) =", p1, "post state =", psi1)
```

von Neumann entropy from eigenvalues (2x2)

For a 2×2 density matrix, eigenvalues can be computed analytically.

```
import math, cmath

def eigvals_2x2(rho):
    a, b = rho[0]
    c, d = rho[1]
    tr = a + d
    det = a*d - b*c
    disc = tr*tr - 4*det
    root = cmath.sqrt(disc)
    lam1 = 0.5*(tr + root)
    lam2 = 0.5*(tr - root)
    # return real parts (should be real for Hermitian rho)
    return [lam1.real, lam2.real]

def S_vn_from_eigs(lams):
    S = 0.0
    for lam in lams:
        lam = max(lam, 0.0)
        if lam > 1e-15:
            S -= lam*math.log(lam, 2)
    return S

rhoA = [[0.5, 0.0],
        [0.0, 0.5]]
print("eigvals:", eigvals_2x2(rhoA))
print("S(rhoA):", S_vn_from_eigs(eigvals_2x2(rhoA)))
```

Build Bell states and compute entanglement entropy

We compute $\rho_A = \mathrm{Tr}_B(\rho_{AB})$ for Bell states and evaluate $S(\rho_A)$.

```
import math

def outer(v):
    n = len(v)
    return [[v[i]*v[j].conjugate() for j in range(n)] for i in range(n

def partial_trace_B(rho):
    # 2 qubits, basis: |00>,|01>,|10>,|11>
    rhoA = [[0j, 0j],[0j, 0j]]
    for a in (0,1):
        for c in (0,1):
            s = 0j
            for b in (0,1):
                i = 2*a + b
                j = 2*c + b
                s += rho[i][j]
            rhoA[a][c] = s
    return rhoA

# Bell states as length-4 vectors
inv = 1/math.sqrt(2)
Phi_plus  = [inv, 0j, 0j, inv]
Phi_minus = [inv, 0j, 0j, -inv]
Psi_plus  = [0j, inv, inv, 0j]
Psi_minus = [0j, inv, -inv, 0j]

for name, psi in [("Phi+", Phi_plus), ("Phi-", Phi_minus),
                  ("Psi+", Psi_plus), ("Psi-", Psi_minus)]:
```

Partially entangled state $\cos\theta|00\rangle + \sin\theta|11\rangle$

This illustrates how entanglement entropy varies continuously from 0 to 1.

```
import math

def psi_theta(theta):
    return [math.cos(theta), 0j, 0j, math.sin(theta)]

print("theta   S(rhoA)")
for k in range(11):
    theta = (math.pi/2) * k/10
    psi = psi_theta(theta)
    rhoAB = outer(psi)
    rhoA = partial_trace_B(rhoAB)
    rhoA_real = [[rhoA[0][0].real, rhoA[0][1].real],
                 [rhoA[1][0].real, rhoA[1][1].real]]
    S = S_vn_from_eigs(eigvals_2x2(rhoA_real))
    print(f"{theta:6.3f}  {S:8.6f}")
```

Check: $S = 0$ at $\theta = 0$ and $\theta = \pi/2$, and $S = 1$ at $\theta = \pi/4$.

Using NumPy

```
import numpy as np

def vn_entropy(rho):
    # eigenvalues of Hermitian matrix
    w = np.linalg.eigvalsh(rho)
    w = np.maximum(w, 0.0)
    w = w[w > 1e-15]
    return -np.sum(w*np.log2(w))

# Bell state |Phi+>
psi = np.array([1,0,0,1], dtype=complex)/np.sqrt(2)
rhoAB = np.outer(psi, psi.conj())

# partial trace over B
rhoA = np.zeros((2,2), dtype=complex)
for a in [0,1]:
    for c in [0,1]:
        for b in [0,1]:
            rhoA[a,c] += rhoAB[2*a+b, 2*c+b]

print(vn_entropy(rhoA))  # should print 1.0
```

Summary

- Shannon entropy $H(X)$ measures classical uncertainty.
- Von Neumann entropy $S(\rho)$ measures quantum mixedness.
- For a pure bipartite state, $S(\rho_A)$ quantifies **entanglement**.
- Bell states have reduced density matrix $\rho_A = I/2$ and entanglement entropy $S = 1$ bit.